

AKSHAY KADAM

Senior AI/ML & Data Engineer | PySpark | AWS | ETL/ELT | Data Lakes | Big Data | Distributed Systems | AI/GenAI

Actively exploring new opportunities in Senior Data Engineering and cloud data platform roles

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PROFESSIONAL SUMMARY

Senior AI/ML & Data Engineer with **6+ years of experience** in data engineering, big data processing, **ETL/ELT pipeline development**, cloud data platforms, and distributed data systems. Experienced in processing real-time and batch vehicle data across the full lifecycle (manufacturing to after-sales). Strong expertise in PySpark, SQL, Python, AWS, Kafka, Airflow, Argo Workflows, HDFS, Hive, data lakes, data warehousing, data modeling, workflow orchestration using Airflow and Argo, CI/CD, and production support. Experienced in building batch and streaming pipelines, integrating data from MySQL and file-based sources, and delivering scalable analytics solutions for enterprise and automotive domains. Hands-on experience in developing AI/GenAI applications using Large Language Models (LLMs) and embedding models for semantic search, natural language querying, and intelligent automation over enterprise data.

CORE SKILLS

Data Engineering: PySpark, Spark, SQL, ETL, ELT, Data Pipelines, Batch Processing, Stream Processing, Data Lakes, Data Warehousing, Data Modeling, Big Data Processing

Programming and Platforms: Python, Pandas, NumPy, Shell Scripting, AWS, S3, EC2, Redshift, Docker, Kubernetes, OpenShift, Hadoop, HDFS, Hive

Orchestration and AI: Kafka, Airflow, Argo Workflows, Bamboo, GitLab CI/CD, GitHub Actions, JFROG, Superset, Trino, Athena, Machine Learning, Deep Learning, MLflow, LLMs, NLP

PROFESSIONAL EXPERIENCE

Senior Data & AI Engineer | Volkswagen Group Digital Solutions | Pune | Sept 2024 - Present

- Architected and implemented a **multi-zone data lake across raw, processed, and curated layers** based on Medallion Architecture to support large-scale vehicle lifecycle analytics on Invehicle and after sales data like warranty, complaints, CAN diagnosis and vehicle master data.
- Owned and **optimized PySpark ETL pipelines** that ingest, transform, and standardize multi-source datasets for **analytics, reporting** and downstream applications. Worked on file formats like **delta, xml, csv, parquet, excel**.
- Involved in **Optimization** of existing ETL Pipelines through **Partitioning and Resources Tuning**.
- Orchestrated resilient production workflows with **Argo Workflows** and managed enterprise-scale object storage using MinIO on OpenShift. Maintaining all CICD artifacts through **JFROG Artifactory for automated deployments**.
- Integrated **Trino and Apache Superset** to enable low-latency distributed querying and **interactive dashboards** for business users.
- Automated **CI/CD with Bamboo** and **deployed containerized services on OpenShift** while managing secure configurations through platform secrets.
- Involved In Discussions regarding **Data Modeling** for different Data layers in Legacy Architecture for extending extraction interfaces to new data sources in Application.
- Also Worked with Infra Team regarding **Data Governance** for giving access to new users also restricting access for some existing users based on Business needs.
- Developed AI Chart Creator to convert natural language queries into SQL and generate dynamic visualizations.
- Built Dashboard Assistant using embedding-based semantic search for intelligent dashboard navigation.
- Created Charging Intelligence Assistant to analyze vehicle charging data, detect patterns, and identify potential faults (e.g., ECU issues, wiring faults).

Data Engineer | Pravaig Dynamics | Bangalore | Mar 2023 - Sept 2024

- Built end-to-end **real-time** and **batch data pipelines** for **GPS, and video data** also in-vehicle diagnosis data analytics on **CAN bus data**. Enabling downstream datasets for deep learning and machine learning engineers.
- Designed AWS-based streaming architecture using **EC2, S3, Redshift, Spark, Kafka, Airflow, and HDFS** for scalable ingestion, processing, and operational analytics.
- Used **Airflow to schedule**, monitor, and recover data workflows reliably while handling task dependencies and failures gracefully.
- Strengthened security and governance by implementing **SASL_SSL-based Kafka authentication** and authorization controls.
- Improved operational reliability through monitoring, logging, shell scripting automation, and **GitLab CI/CD** for AWS-hosted deployments.

Data Engineer | Xoriant | Pune, Maharashtra | Feb 2021 - Feb 2023

- Analyzed retail-domain business requirements and translated functional and reporting needs into scalable data engineering solutions.
- Ingested and consolidated retail data from MySQL and file-based sources using Apache Spark and **Databricks notebooks**.
- Built batch ETL pipelines using PySpark and SQL, transforming raw retail data into curated **Delta Lake tables** for analytics and downstream consumption.
- Implemented data cleansing, validation, enrichment, and schema management across bronze, silver, and gold data layers.
- Developed and monitored **Databricks workflows** to automate retail data ingestion and transformation processes.

GLOBAL ONSITE EXPERIENCE

Wolfsburg, Germany

- Partnered with global stakeholders to evaluate vehicle data systems, align technical direction, and shape scalable data strategies for enterprise use cases.
- Delivered AI proof-of-concept solutions on real enterprise datasets, securing approval for production deployment.
- Supported the transition of AI/ML solutions from proof-of-concept stages into production-ready systems integrated with enterprise data pipelines.

AWARD & RECOGNITION

Innovation Award 2026 | VWGDS [Link](#)

- Recognized for contributing to innovative data/analytics solutions and product improvements through AI solutions. Specially for CIA (Charging Intelligent Assistance), Dashboard Assistance and Chart Creator.

Wall of Fame Award 2025 | VWGDS [Link](#)

- Recognized for Delivering GDC Data Product for vehicle's diagnostic data demonstrating effective contribution to get insights of Vehicle's health and fleet reporting.

CERTIFICATIONS

[Databricks - Lakehouse](#) | [Databricks - Platform Architect](#) | [Basics of Machine Learning](#) | [Machine Learning Algorithms](#)

EDUCATION

iNeuron AI, Bangalore, Karnataka | *Masters of Data Science* | May 2023 - May 2024

ZCOER, Pune, Maharashtra | *Bachelor of Engineering* | Jun 2012 - Jun 2016